Neural Dynamics of Face Perception Under Visual Noise: Investigating High-Gamma and Broadband Responses in Perceived and Non-Perceived Faces Using ECoG Dataset

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Introduction

- **Face Perception**
  The brain can detect and recognize faces even when visual information is degraded by noise.

- **Visual Noise**
  Increasing visual noise reduces the available facial information and makes face perception more challenging.

- **ECoG**
  Electrocorticography (ECoG) provides high-temporal-resolution recordings of cortical activity, allowing us to study the neural dynamics of face perception.

- **Neural Responses**
  High-gamma, Beta and broadband activity provide sensitive measures of cortical processing. Their amplitude and latency can reveal differences in neural responses between perceived and non-perceived faces under different levels of visual noise.

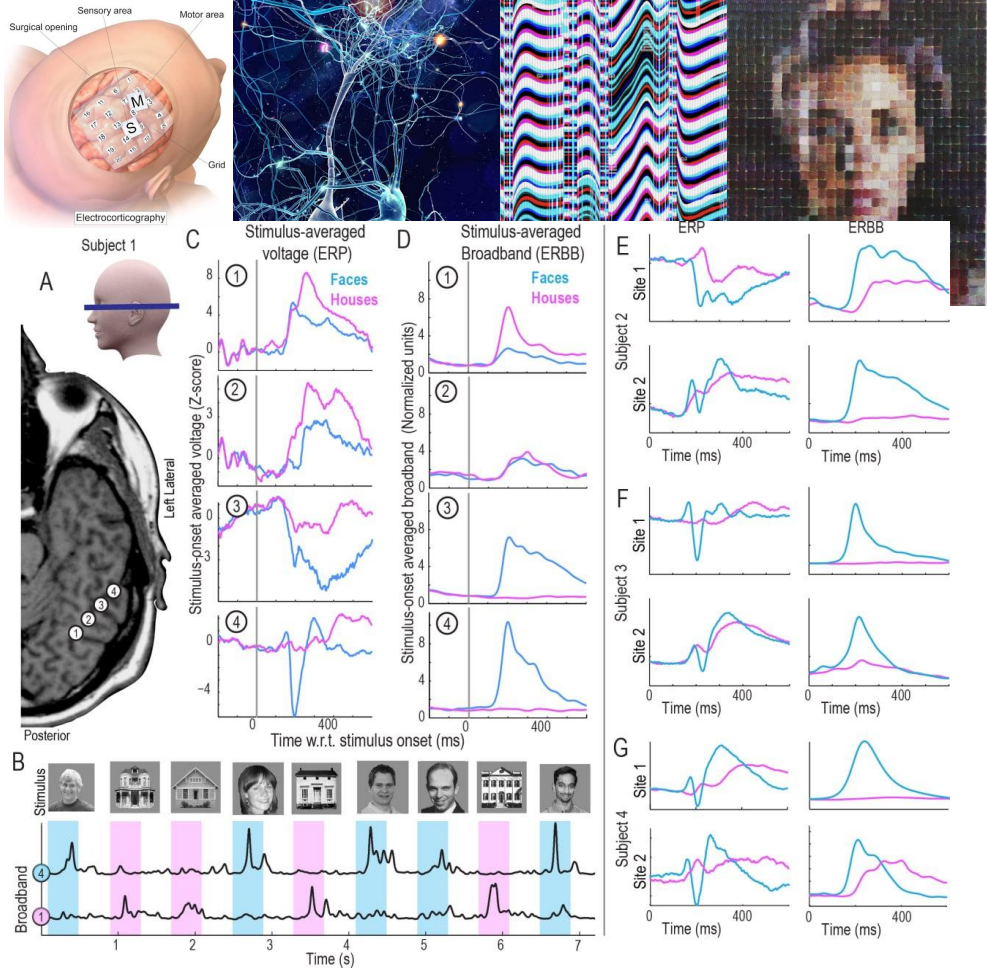


Fig 1. The basic face and house discrimination task, and the polymorphic nature of the electrophysiologic response.

Goals

Investigate neural response amplitude and latency during face perception under different levels of visual noise.

Compare perceived (Hit) and non-perceived (Miss) trials using high-gamma, Beta and broadband activity at face-selective electrodes.

Evaluate whether neural activity can distinguish perceived from non-perceived faces across noise levels.

The Neuromatch ECoG Faces/Houses Dataset

7 Subjects, Two Task

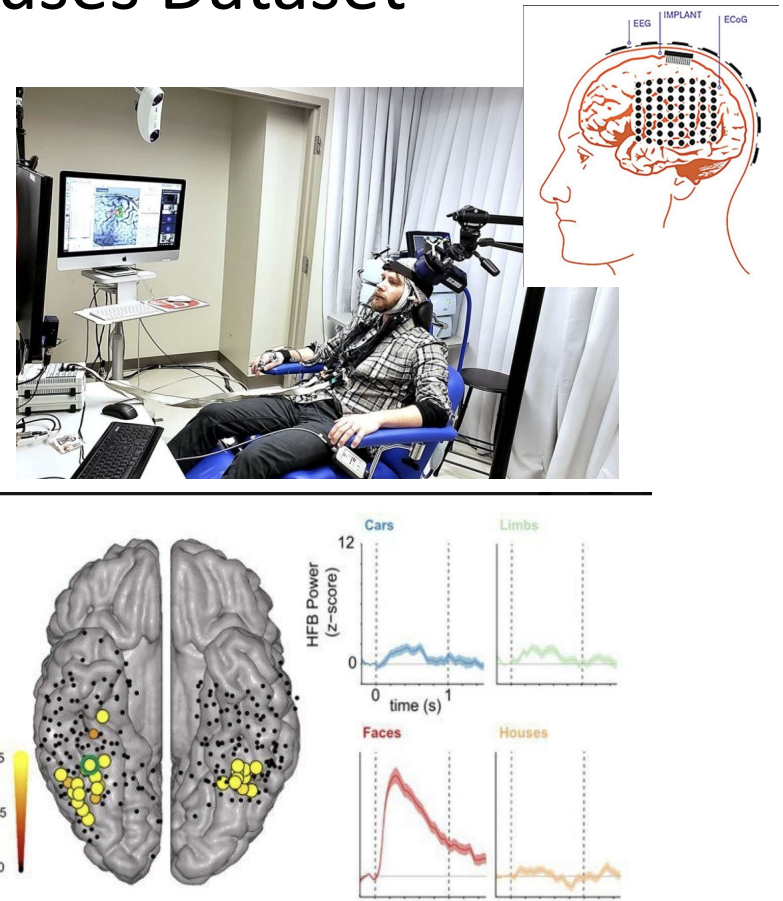
ECoG recordings from 7 patients: a localizer identifies face-selective electrodes, then a noisy task tests perception at 0–100% noise.

Behavioral Response

A button press marks each trial as Hit (perceived) or Miss (non-perceived).

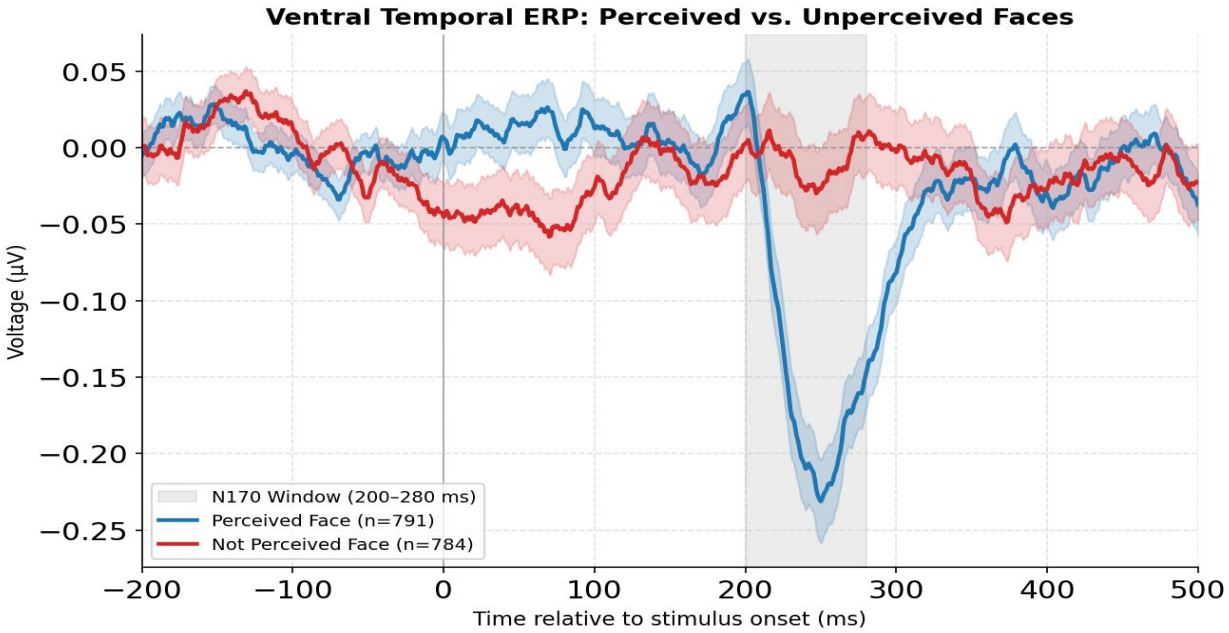
Neural Measures

High-gamma and broadband activity are used to measure response amplitude and latency.



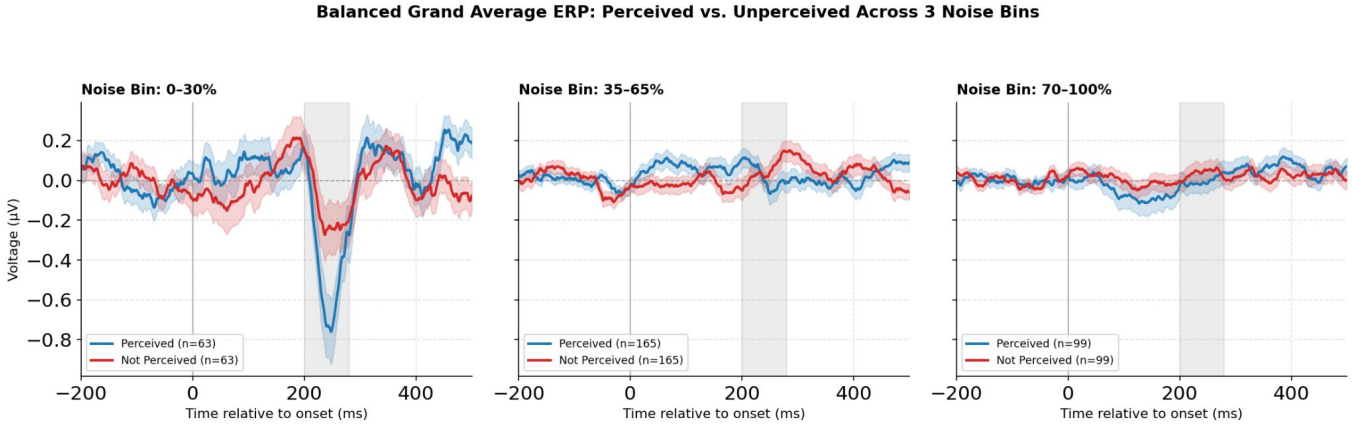
Results: ERP & Spectral Responses

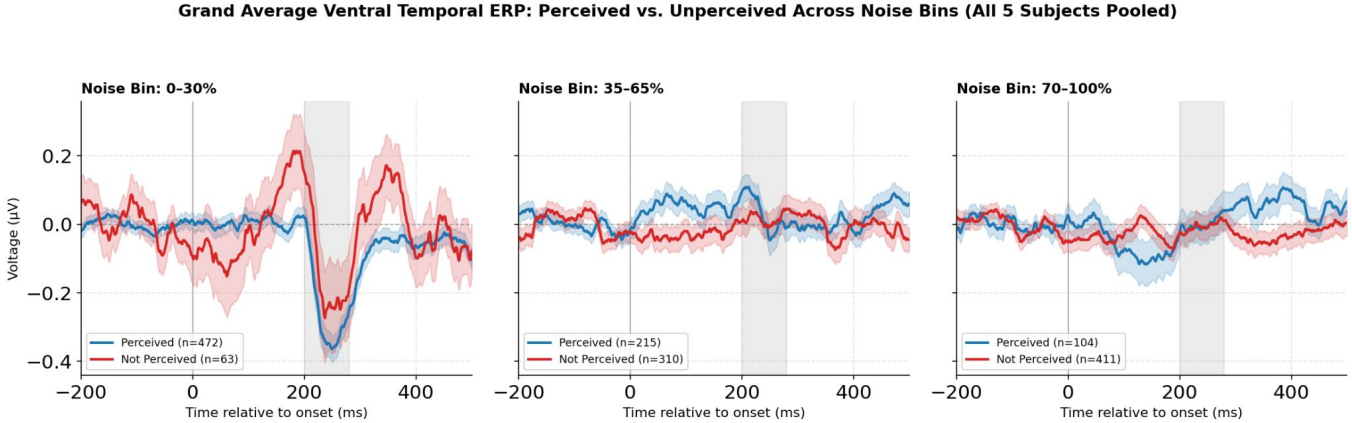
- Perceived faces show a much deeper N170 dip than non-perceived faces.
- The gap emerges sharply within the 200–280ms window.



Results: ERP & Spectral Responses

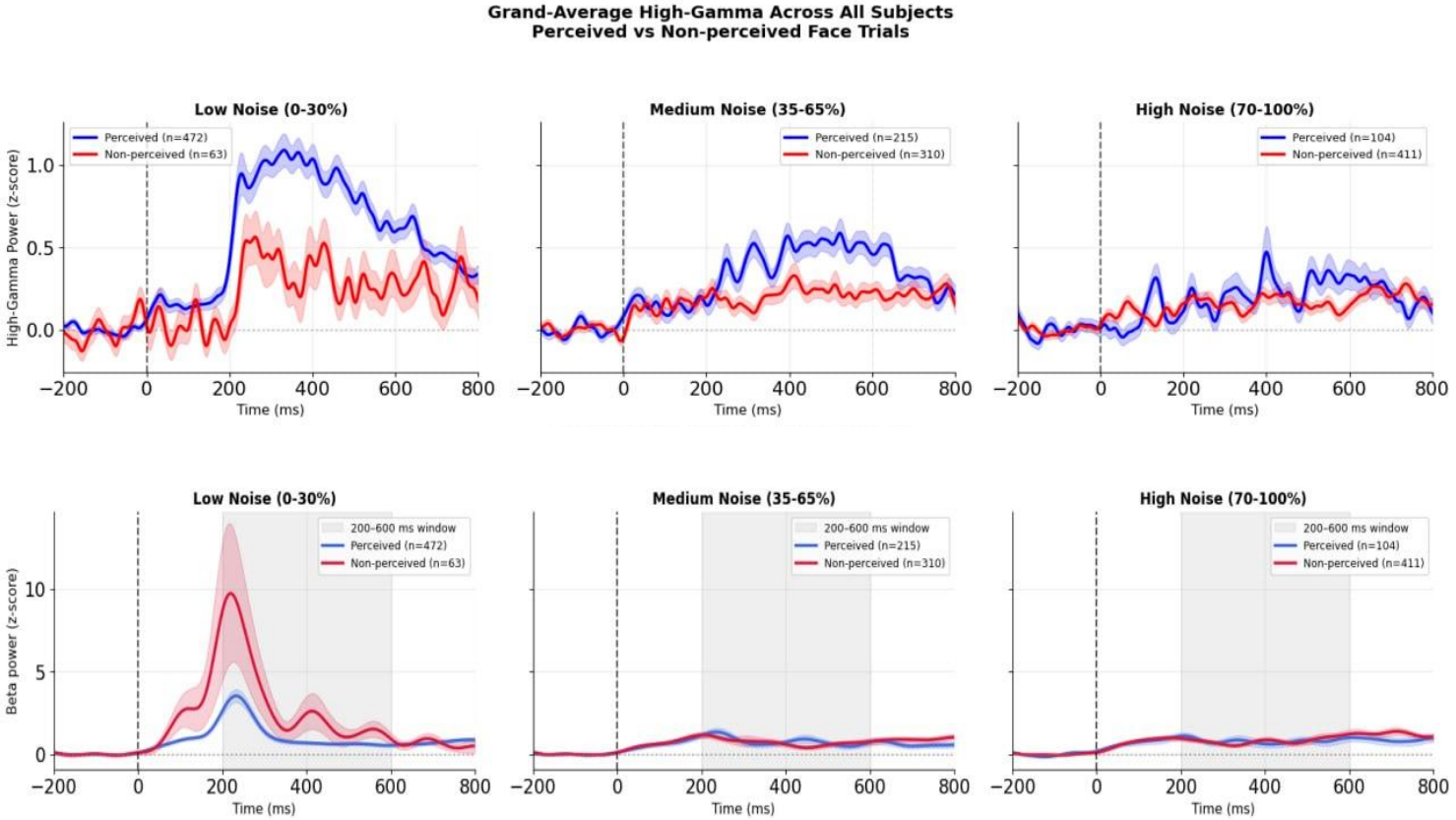
- Low noise shows the strongest divergence between perceived and non-perceived responses.
- This N170 gap shrinks noticeably as visual noise increases.
  At high noise, perceived and non-perceived traces nearly overlap.
- Balanced and pooled averaging methods produce highly consistent overall patterns.
- This supports a noise-dependent perceptual threshold in face-selective regions.



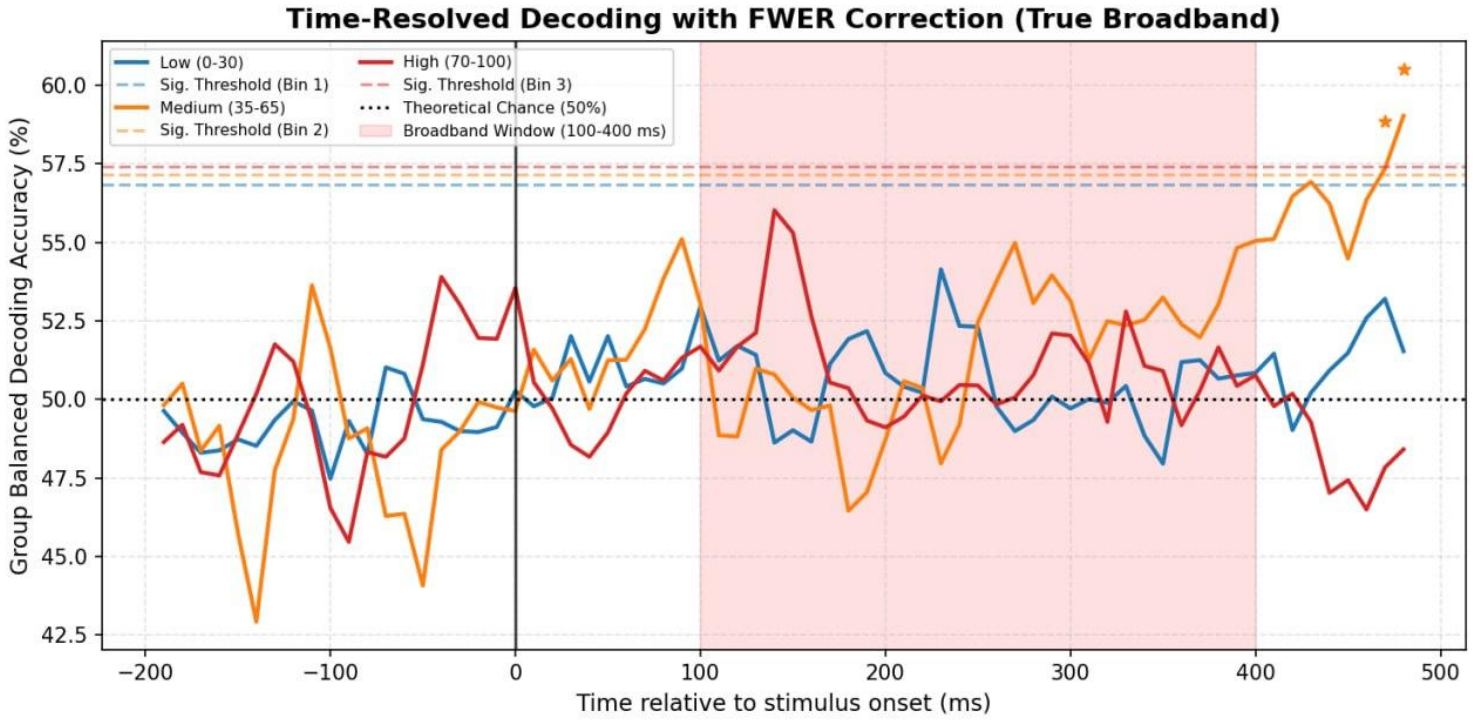


Results: High-Gamma Vs. Beta

- High-Gamma power was greater for Perceived vs. Non-perceived trials, especially low-medium noise.
- Both effects diminished at high noise, where perception and neural response converge.
- Beta showed the opposite pattern—Non-perceived trials had stronger Beta responses at low noise

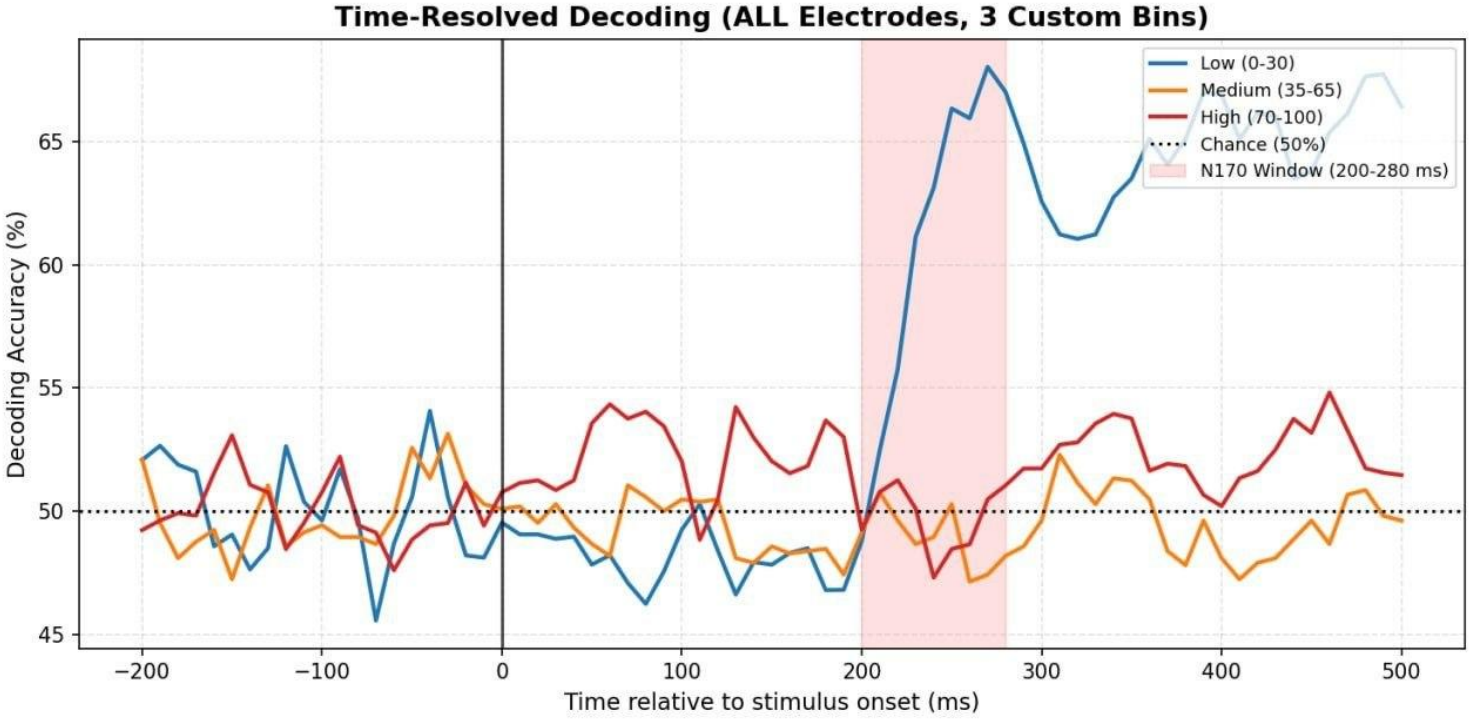


Results: Perceived & Non-perceived LDA



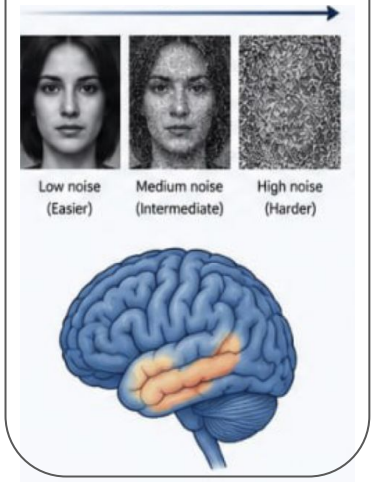
Results: Broadband LDA

- Accuracy climbs sharply within the N170 window.
- Medium and high noise stay near chance.

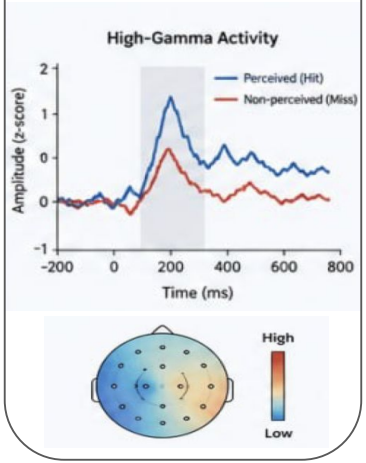


Applications

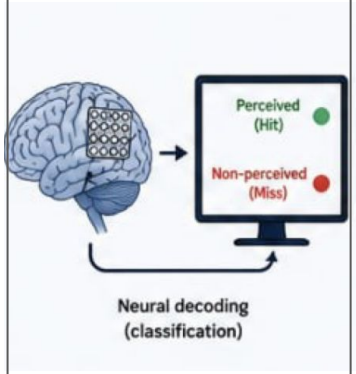
1.Understanding visual perception under noisy condition



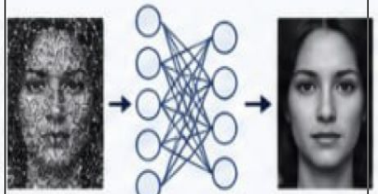
2.Identifying neural markers of conscious perception



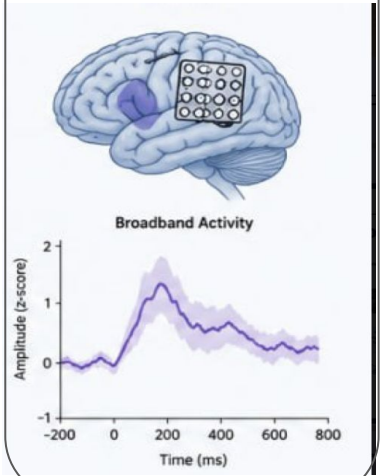
3.Brain-Computer Interfaces And neural decoding



4.Robust visual recognition inspired by neural processing



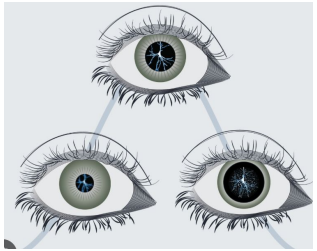
5.Advancing cognitive neuroscience research

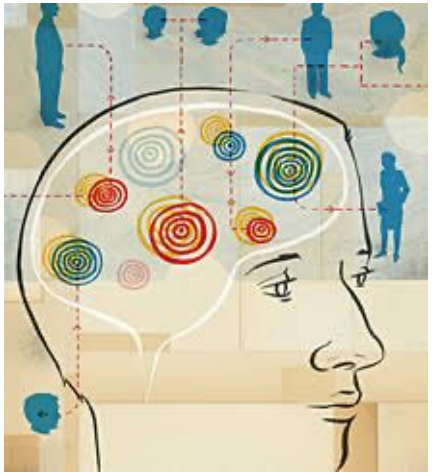


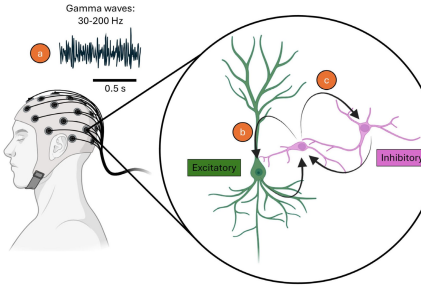
Discussion

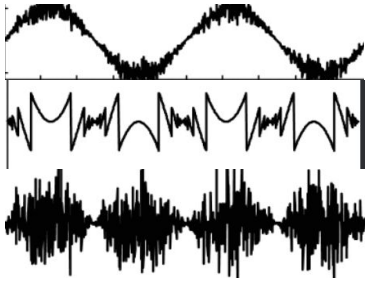
- **Noise Weakens the Distinction**
  As noise increases, perceived and non-perceived responses become harder to distinguish.

- **Beyond Physical Input**
  Neural activity reflects not just stimulus quality, but conscious perceptual awareness.









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